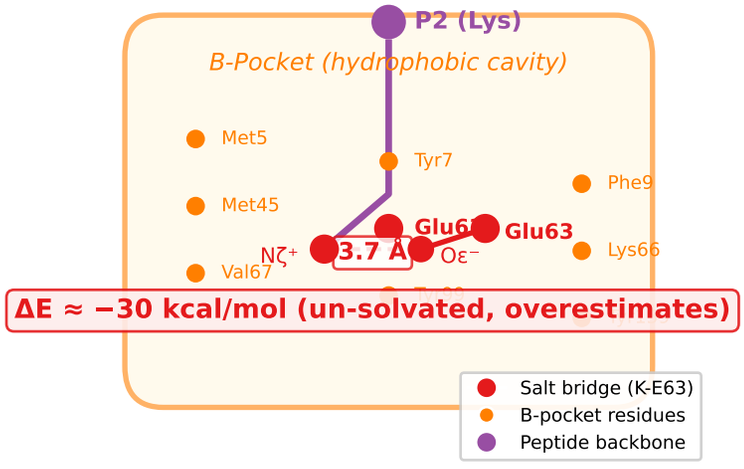
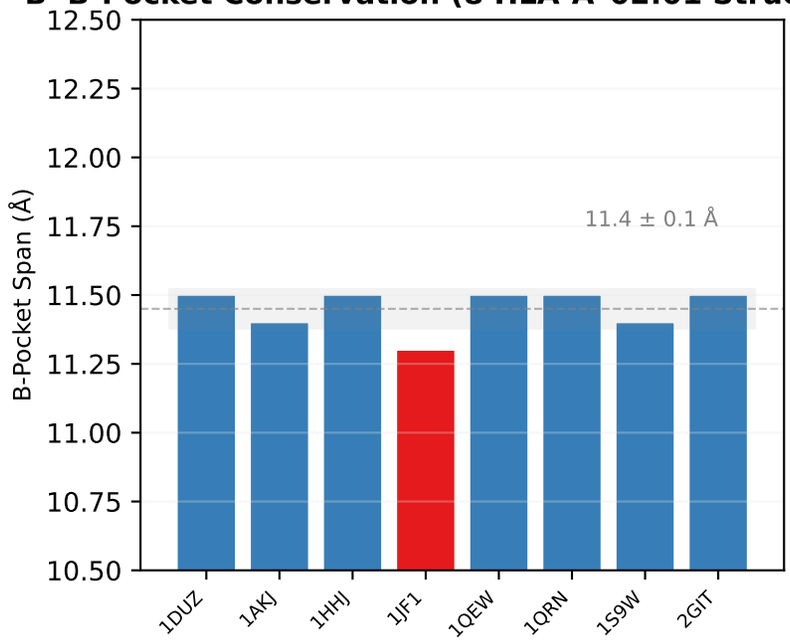
K-E63 Salt Bridge Structural Hypothesis (KRAS G12V YKLVVVGAV)

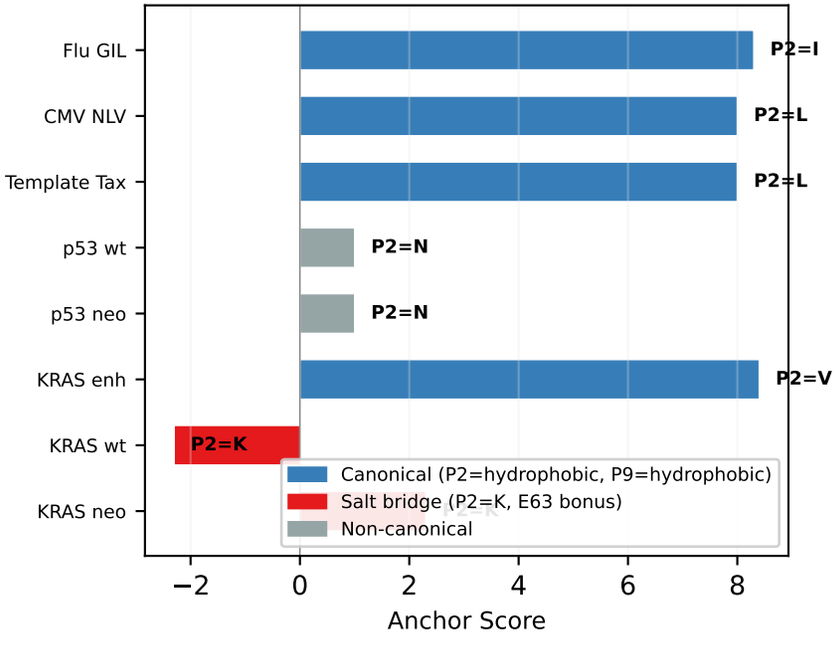
A B-Pocket Salt Bridge: K(P2) → Glu63



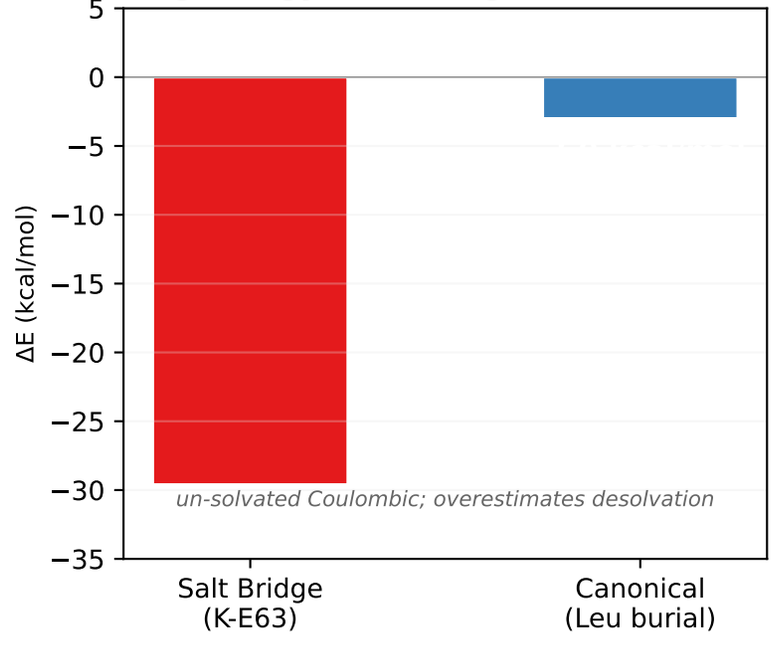
B B-Pocket Conservation (8 HLA-A*02:01 Structures)



C Peptide Anchor Scores (P2 + P9 Motif)



D Binding Energy: Salt Bridge vs Canonical Anchor



E K-E63 Salt Bridge — Structural Evidence Summary

Method:	pdbfixer + OpenMM minimization (amber14, no cutoff)
Template:	1DUZ (HLA-A*02:01 + LLFGYPVYV, 1.80 Å)
P2 CA → Glu63 OE2:	3.7 Å
P2 CA → Glu63 OE1:	3.9 Å
Min OE distance:	3.7 Å
K sidechain reach (CA→Nz):	7.6 Å
Salt bridge cutoff:	4.0 Å (standard N–O definition)
Coulomb ΔE (salt bridge, un-solvated):	-29.6 kcal/mol (overestimates)
Canonical Leu burial ΔG:	−3.0 kcal/mol (hydrophobic anchor)
B-pocket conservation (8 PDBs):	11.4 ± 0.1 Å
Verdict:	**Qualitative hypothesis — not quantified**

Qualitative hypothesis only; pending solvated free-energy calculation and experimental validation